

Gaspar Ndokaj

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EDUCATION

Vassar College

Bachelor of Arts in Mathematics, Minor in Continental Philosophy

Poughkeepsie, NY

Sep. 2024 – May 2028

University of Arizona (Southwest Center for Arithmetic Geometry)

Non-degree seeking student

Tucson, AZ

Sep. 2026 - Nov. 2026

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Polymath Jr.

Jun. 2026 – Present

Remote

- Performed collaborative research on Binomial Edge Ideals under the supervision of Prof A. Seceleanu from the University of Nebraska-Lincoln.

Undergraduate Research Assistant

Vassar College

Sep. 2025 – May 2026

Poughkeepsie, NY

- Quantizing chaotic markers of dynamic diffraction time-series data of *C. elegans* locomotion in 3D using Recurrence Quantification Analysis, Lyapunov exponents, and Information Theory.
- Utilized techniques of nonlinear time-series analysis such as Takens's Embedding Theorem and Kolmogorov-Sinai entropy in MATLAB to show chaotic markers.

INVITED TALKS/CONFERENCES

Hudson River Undergraduate Mathematics Conference (HRUMC) | *Speaker*

Apr. 2026

- Gave a talk on Hutchinson's theorem and Iterated Function Systems in Fractal Geometry with Jay Chen and Henry Bonney at the HRUMC conference, under the guidance of Professor Natalie Frank of Vassar College.
- Gave intuition and background behind Hutchinson's theorem and focused on the topological and fractal geometric aspects of iterated function systems and Hutchinson's theorem for the existence and uniqueness of fractals.

Polymath Jr. Online Conference | *Speaker*

Aug. 2026

- Gave a talk on Binomial Edge Ideals alongside Shantanu Chakraborty of Clemson University, Ilya Galushkin of University of Pennsylvania, and Dragos Gilgor of Babes-Bolyai University for the ending conference for the Polymath Jr. Program.
- Focused on the Weak and Strong Lefschetz Properties for graded Artinian k -algebras and talked about findings from the Polymath Jr. program on their implications for binomial edge ideals of families of graphs.

RELEVANT COURSEWORK

Vassar College

Sep. 2024 – Present

MATH125, Topics in Single Variable Calculus

MATH220, Multivariable Calculus

MATH221, Linear Algebra

MATH228, Ordinary Differential Equations and Applications

MATH263, Discrete Mathematics

MATH261, Introduction to Number Theory

MATH321, Real Analysis

MATH361, Modern Algebra

MATH298, Independent Work (Computational Algebraic Geometry)

CMPU240, Theory of Computation

CMPU102, Data Structures and Algorithms

University of Arizona (Southwest Center for Arithmetic Geometry)

Preliminary Arizona Winter School, Algebraic Geometry via Computation

Sep. 2026 – Nov. 2026

TECHNICAL SKILLS

Languages: Java, $\text{\LaTeX}$, Wolfram Mathematica, Macaulay2, MATLAB, OCaml, Lua

Developer Tools: Git, Docker, Bash/zsh, Vim, Neovim, VS Code, IntelliJ, Debian Linux, Arch Linux, apt, pacman

Languages: English (Fluent), Albanian (Fluent), Japanese (Elementary)